CURRICULUM VITAE of LONGHAI LI (Jul 26, 2026)

1. PERSONAL

- Official webpage: https://artsandscience.usask.ca/profile/LLi
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- Phone: +1 (306) 966-6095
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  Department of Mathematics & Statistics
  University of Saskatchewan
  106 Wiggins RD
  Saskatoon, SK, S7W0G8 CANADA

2. DEGREES

- Ph.D., University of Toronto, 2007, Statistics
  Supervisor: Radford M. Neal
  Thesis: Bayesian Classification and Regression with High Dimensional Features
- M.Sc., University of Toronto, 2003, Statistics
- B.Sc., University of Science and Technology of China, 2002, Statistics.

4. Employment History

- Full Professor, July 1st, 2018, Dept. of Math & Stat., Univ. of Saskatchewan, SK, Canada
- Associate Professor, July 1st, 2012, Dept. of Math & Stat., Univ. of Saskatchewan, SK, Canada
- Assistant Professor, July 1st, 2007, Dept. of Math & Stat., Univ. of Saskatchewan, SK, Canada
- Research Intern, Nov. 2006 to Feb. 2007, Microsoft Research, Redmond, WA, USA
- Sessional Instructor, 2006-2007, University of Toronto, Toronto, ON, Canada

7. LEAVES

- Sabbatical Leave, Jan. 1st, 2025 to June 30th, 2025.
- Sabbatical Leave, July 1st, 2020 to June 30th, 2021.
- Parental Leave, Jan. 1st, 2019 to June 30th, 2019.
- Sabbatical Leave, July 1st, 2013 to June 30th, 2014.

9. TEACHING ACTIVITIES

9.1 Scheduled Instructional Activity

2025-2026

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 851	T2	Linear Statistical Models	LEC	4	39	156
STAT 443	T2	Linear Models	LEC	3	39	117
STAT 442	T2	Statistical Inference	LEC	3	39	117
STAT 850	T2	Mathematical Statistics and Inference	LEC	1	39	39

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 845	T1	Statistical Methods for Research	LEC	10	39	390
MATH 996	T1T2	Research Supervision (Ph.D.)	RES	1		
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	1		

2024-2025

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 348	T1	Sampling Techniques	LEC	15	39	585
STAT 812	T1	Computational Statistics	LEC	5	39	195
STAT 420	T1	Topics in Computational Statistics	LEC	1	39	39
BIOS 996	T1T2	Research Supervision (Ph.D.)	RES	1		
BIOS 996	T1T2	Research Supervision (M.Sc.)	RES	1		
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	1		

2023-2024

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 245	T1	Introduction to Statistical Methods	LEC	91	39	3549
STAT 812	T1	Computational Statistics	LEC	5	39	195
STAT 420	T1	Topics in Computational Statistics	LEC	1	39	39
STAT 443	T2	Linear Statistical Models	LEC	4	39	156
STAT 851	T2	Linear Models	LEC	4	39	156
BIOS 996	T1T2	Research Supervision (Ph.D.)	RES	1		
BIOS 996	T1T2	Research Supervision (M.Sc.)	RES	1		
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	1		

2022-2023

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 348	T1	Sampling Techniques	LEC	35	39	1365
STAT 812	T1	Computational Statistics	LEC	3	39	117
STAT 420	T1	Topics in Computational Statistics	LEC	1	39	39
STAT 443	T2	Linear Statistical Models	LEC	1	39	39
STAT 851	T2	Linear Models	LEC	1	39	39
BIOS 996	T1T2	Research Supervision (M.Sc.)	RES	1		
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	1		
BIOS 996	T1T2	Research Supervision (Ph.D.)	RES	1		

2021-2022

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 244	T2	Elementary Statistical Concepts	LEC	72	39	2808
STAT 342	T1	Mathematical Statistics	LEC	8	39	312
STAT 443	T2	Linear Statistical Models	LEC	3	39	117
STAT 851	T2	Linear Models	LEC	5	39	195
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	2		
MATH 994	T1T2	Research Supervision (Ph.D.)	RES	1		
BIOS 996	T1T2	Research Supervision (Ph.D.)	RES	1		
ENGR 996	T1T2	Research Supervision (Ph.D.)	RES	1		

2020-2021

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	3		
BIOS 996	T1T2	Research Supervision (Ph.D.)	RES	1		
ENGR 996	T1T2	Research Supervision (Ph.D.)	RES	1		

2019-2020

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 245	T1	Introduction to Statistical Methods	LEC	128	39	4992
STAT 345	T2	Design and Analysis of Experiments	LEC	27	39	1053
STAT 834	T2	Advanced Experimental Design	LEC	1	39	39
STAT 812	T1	Computational Statistics	LEC	5	39	195
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	1		
BIOS 996	T1T2	Research Supervision (Ph.D.)	RES	1		
ENGR 996	T1T2	Research Supervision (Ph.D.)	RES	1		

2018-2019

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 812	T1	Computational Statistics	LEC	11	39	429
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	2		
BIOS 994	T1T2	Research Supervision (M.Sc.)	RES	2		
BIOS 994	T2	Research Supervision (Ph.D.)	RES	1		
MATH 996	T1T2	Research Supervision (Ph.D.)	RES	1		

2017-2018

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 241	T1	Probability Theory	LEC	70	39	2730
STAT 345	T2	Design and Analysis of Experiments	LEC	25	39	975
STAT 834	T2	Advanced Experimental Design	LEC	9	39	351
STAT 841	T2	Probability Theory	LEC	5	39	195
MATH 994	T1T2	M.Sc. Research Supervision	RES	5		

2016-2017

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 812	T1	Computational Statistics	LEC	12	39	468
STAT 348	T2	Sampling Techniques	LEC	33	39	1287
STAT 442	T2	Statistical Inference	LEC	1	39	39
STAT 846	T2	Sp. Topics (Statistical Inference)	LEC	8	39	312
MATH 994	T1T2	M.Sc. Research Supervision	RES	6		

2015-2016

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 245	T1	Intro. to Stat. Methods	LEC	157	39	6123
STAT 342	T1	Mathematical Statistics	LEC	7	39	273
STAT 242	T2	Stat. Theory & Methodology	LEC	11	39	429
STAT 245	Summer	Intro to Statistical Methods	LEC	30	39	1170
MATH 994	T1T2	M.Sc. Research Supervision	RES	5		
MATH 996	T1T2	Ph.D. Research Supervision	RES	1		

2014-2015

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 342	T1	Mathematical Statistics	LEC	9	39	351
STAT 812	T1	Computational Statistics	LEC	10	39	390
STAT 348	T2	Sampling Techniques	LEC	21	39	819
STAT 442	T2	Statistical Inference	LEC	5	39	195
STAT 846	T2	Sp. Topics (Statistical Inference)	LEC	8	39	312
STAT 244	T2	Elementary Statistical Concepts	LEC	14	39	546
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	4		
MATH 996	T1	Research Supervision (Ph.D.)	RES	2		

2012-2013

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 241	T1	Probability Theory	LEC	51	39	1989
STAT 348	T2	Sampling Techniques	LEC	15	39	585
STAT 245	T2	Intro to Statistical Methods	LEC	135	39	5265
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	2		
MATH 996	T1T2	Research Supervision (Ph.D.)	RES	2		

2011-2012

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 241	T1	Probability Theory	LEC	43	39	1677
STAT 342	T1	Mathematical Statistics	LEC	3	39	117
STAT 841	T1	Probability Theory	LEC	10	39	390
STAT 245	T2	Intro to Statistical Methods	LEC	138	39	5382
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	2		

2010-2011

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 241	T1	Probability Theory	LEC	42	39	1638
STAT 846	T1	Computational Statistics	LEC	8	39	195
STAT 242	T2	Stat. Theory & Methodology	LEC	13	39	507
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	2		
MATH 996	T1T2	Research Supervision (Ph.D.)	RES	1		

2009-2010

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 342	T1	Mathematical Statistics	LEC	4	39	156
STAT 841	T1	Probability Theory	LEC	6	39	234
STAT 241	T2	Probability Theory	LEC	28	39	1092
STAT 244	T2	Elem. Stat. Concepts	LEC	83	39	3237
STAT 848	T2	Multivariate Data Analysis	READ	2	39	78
MATH 994	T1T2	Research Supervision (M.Sc.)	RES	2		
MATH 996	T1T2	Research Supervision (Ph.D.)	RES	1		

2008-2009

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 342	T1	Mathematical Statistics	LEC	19	39	741

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 848	T2	Multivariate Data Analysis	LEC	6	39	234

2007-2008

COURSE	TERM	TITLE	TYPE	ENRL	YIH	YCSH
STAT 342	T1	Mathematical Statistics	LEC	7	39	273
STAT 846	T2	Computational Statistics	LEC	5	39	195

9.2 Unscheduled Instructional Activity

- Creating and assessing PhD qualifying exam on Mathematical Statistics, July 2021.
- Creating and assessing PhD qualifying exam on Mathematical Statistics, May 2021.

9.3 Course and Program Development

2022-2023

[3] Renewal of the University of Saskatchewan Courses for SSC Accreditation.

[2] Participated in creating new M.Sc. and Ph.D. programs in Statistics, approved on May 18, 2023.

2020-2021

[1] Participation in creating Certificate in Statistical Methods, 2021.

9.4 Teaching Materials

2025-2026

[4] STAT 442/850: a textbook entitled "Theory of Statistical Inference" in PDF ($\geq$ 200 pages) and HTML.

[3] STAT 845: a textbook entitled "Statistical Inference and Learning Methods for Research" in PDF ($\geq$ 200 pages) and HTML

[2] STAT 443/851: a textbook entitled "Theory of Linear Model" in PDF ($\approx$ 200 pages) and HTML.

2024-2025

[1] STAT 443/851: 273 pages of handwritten lecture notes were developed and posted to students.

9.5 Other Teaching-Related Activities

2025-2026

[10] Peer evaluation for Matthew Schmirler, Mar. 2026.

2023-2024

[9] Data Science Bootcamp, a case study, June 19, 2023.

2022-2023

[8] Peer evaluation for Raj Srinivasan, April 2023.

[7] Peer evaluation for Saima Khosa, Dec. 2022.

2021-2022

[6] Peer Teaching Evaluation for Shahedul Khan, Nov. 2021.

[5] Peer Teaching Evaluation for Li Xing, Nov. 2021.

2020-2021

[4] Peer Teaching Evaluation for Shahedul Khan, STAT 812, Nov. 12, 2020.

2019-2020

[3] Peer Teaching Evaluation for Annalizer McGillvray, Nov. 29, 2019.

2017-2018

[2] Peer Teaching Evaluation for Shahedul Khan, 2017-2018.

[1] Peer Teaching Evaluation for Lawrence Chang, 2017-2018.

10. SUPERVISION AND ADVISORY ACTIVITIES

10.1 Undergraduate Student Supervision

2025-2026

[16] George Chen, B.Sc., Computer Science, Simon Fraser University, June 1, 2025 – Aug. 31, 2025, Supervised.

[15] Shruti Kaur, B.Sc., Computer Science and Statistics, May – Aug., 2025, Supervised.

2023-2024

[14] George Chen, B.Sc., Computer Science, Simon Fraser University, May 7, 2023 – Aug. 25, 2023, Supervised.

2022-2023

[13] Noah Little, B.Sc., Anatomy and Cell Biology, April 2022 – Aug. 31, 2022, Supervised.

2020-2021

[12] Noah Little, B.Sc., Anatomy and Cell Biology, Aug. 2020 – Feb. 2021, Supervised.

[11] Lifang Lei, B.Sc., Mechanical Engineering, Oct. 2020 – March 2021, Supervised.

[10] Yanping Li, B.Sc., Business, Edward Business School, Feb. 2021 – June 2021, Supervised.

[9] Lina Li, B.Sc., Statistics, May 2020 – June 2020, Summer Research Assistant, Supervised.

2019-2020

[8] Hao Hu, B.Sc., Statistics, 2019–2020, Undergraduate Thesis Supervision, Supervised.

[7] Steven Liu, B.Sc., Computer Science, July 2019 – Oct. 2020, Summer Research Assistant, Supervised.

- Project title: Development of an R package for Bayesian Hyper-LASSO Logistic Regression
- Software: HTLR: Bayesian Logistic Regression with Heavy-Tailed Priors, **[CRAN page]**, **[Github page]**. HTLR was listed in the top 40 new packages in October 2019 by R views of Rstudio.

[6] Steven Liu, B.Sc., Computer Science, June 17, 2019 – Aug. 31, 2019, Supervised.

2018-2019

[5] Jian Su, B.Sc., Computer Science, May 1, 2018 – Aug. 31, 2018, Supervised.

2016-2017

[4] Jiaqi Xiao, B.Sc., Economics, Research Assistant, May 2016 – Aug. 2016, Supervised.

2015-2016

[3] Jiaqi Xiao, B.Sc., Economics, May 2015 – Aug. 2015, Supervised.

2014-2015

[2] Zhouji Zhang, B.Sc., Mathematics, June 1, 2014 – June 30, 2014, Supervised.

2013-2014

[1] Bei Zhang, B.Sc., Statistics, May 1, 2013 – Aug. 31, 2013, Supervised.

10.2 Graduate Student Supervision

Ph.D. Students

[4] **Jing Wang**, Ph.D., Biostatistics, School of Public Health, Co-supervised with Prof. Li Xing, 2023–2026 (Transferred to other supervisor)

[3] **Wutao Yin**, Ph.D., Biomedical Engineering, Co-supervised with Prof. FangXiang Wu, 2019–2021 (Defended: Dec. 17, 2021)

- **Thesis:** Artificial Intelligence Based Methods for Autism Spectrum Disorder Diagnosis from fMRI Data
- **Employment:** Associate Professor, Ocean Institute, Northwestern Polytechnical University, Taicang Jiangsu, China

[2] **Tingxuan Wu**, Ph.D., Biostatistics, School of Public Health, Co-supervised with Prof. Cindy Feng, 2018–2023 (Defended: May 24, 2023)

- **Thesis:** Residual Diagnostics and Statistical Inference for Shared Frailty Models
- **Employment:** Forestry Statistical Analyst, Ministry of Environment, Government of Saskatchewan

[1] **Lai Jiang**, Ph.D., Statistics, Math & Stat, Supervised, 2009–2015 (Defended: Sept. 14, 2015)

- **Thesis:** Fully Bayesian T-probit Regression with Heavy-tailed Priors for Selection in High-Dimensional Features with Grouping Structure
- **Employment:** Postdoctoral fellow at Lady Davis Institute, Jewish General Hospital, McGill University in Montreal

Master's Students

[18] **Dananji Egodage (Shashiprabha)**, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Cindy Feng, 2023–2025 (Defended: Aug. 30, 2025)

- **Thesis:** Component-wise Z-residual Diagnosis for Bayesian Hurdle Models
- **Employment:** Applied Researcher at Southeast College Saskatchewan

[17] **Wuqian Effie Gao**, M.Sc., Biostatistics, School of Public Health, Co-supervised with Prof. Cindy Feng, 2022–2024 (Defended: Aug. 30, 2024)

- **Thesis:** Z-residuals for Checking Bayesian Hurdle Models
- **Employment:** Senior Data and Forecasting Analyst, Health Ministry, Government of Saskatchewan

[16] **Lina Li**, M.Sc., Statistics, Math & Stat, MITACS Project supervisor, Supervised, 2022–2022 (Completed: August 2022)

[15] **Hao Hu**, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Li Xing, 2021–2022 (Defended: Sept. 15, 2022)

- **Thesis:** Identifying Risk Factors for Cognitive Decline Using Statistical Learning Techniques and Functional Data Analysis
- **Employment:** Senior Statistician in Health Ministry in a Chinese government

[14] **Man Chen**, M.Sc., Statistics, Math & Stat, Supervised, 2019–2021 (Defended: April 30, 2021)

- **Thesis:** Association between Gut Microbiome and Parkinson's Disease Revealed by Sparse Learning
- **Employment:** AI Engineer at Super GeoAI Technology Inc.

[13] **Mei Dong**, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Lloyd Balbuena, 2017–2019 (Defended: May 23, 2019)

- **Thesis:** Feature Selection Bias in Assessing the Predictivity of SNPs for Alzheimer's Disease
- **Employment:** Senior Research Analyst at the University of Toronto

[12] **Tingxuan Wu**, M.Sc., Biostatistics, School of Public Health, Co-supervised with Prof. Cindy Feng, 2017–2018 (Defended: Dec. 4, 2018)

- **Thesis:** Randomized Survival Probability Residuals for Assessing Parametric Survival Models
- **Employment:** PhD student at the University of Saskatchewan

[11] **Xiaoying Wang**, M.Sc., Statistics, Math & Stat, Supervised, 2016–2019 (Defended: March 12, 2019)

- **Thesis:** Comparison of Statistical Testing and Predictive Analysis Methods for Feature Selection in Zero-inflated Microbiome Data

[10] **Wei Bai**, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Cindy Feng, 2016–2018 (Defended: July 12, 2018)

- **Thesis:** Randomized Quantile Residual for Assessing Generalized Linear Mixed Models with Application to Zero-Inflated Microbiome Data
- **Employment:** Lead Statistical Analyst, Bayer Pharmaceuticals, Ontario, Canada (First Employment: Statistical Programmer, Everest Clinical Research, Markham, Ontario)

[9] **Arash Shamloo**, M.MATH., Statistics, Math & Stat, Supervised, 2016–2017 (Project Completed: August 31, 2017)

- **Project:** Randomized quantile residuals for accelerated failure time models
- **Employment:** Research assistant, College of Pharmacy and Nutrition, University of Saskatchewan

[8] **Alireza Sadeghpour**, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Cindy Feng, 2016–2017 (Defended: Sept. 19, 2017)

- **Thesis:** Empirical Investigation of Randomized Quantile Residuals for Diagnosis of Non-Normal Regression Models
- **Employment:** Statistician at Health Canada, Ottawa

[7] **Yunyang Wang**, M.Sc., Statistics, Math & Stat, Supervised, 2014–2017 (Defended: Nov. 18, 2017)

- **Thesis:** Comparison of Stochastic Volatility Models Using Integrated Information Criteria
- **Employment:** Statistician at Montreal office of Evidera (a PPD company), Montreal, QC (First Employment: Intern at the PathWise Solutions, AON Securities, Toronto)

[6] **Naorin Islam**, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Shahedul Khan, 2014–2017 (Defended: Nov. 28, 2017)

- **Thesis:** Substance Abuse and Health: A Structural Equation Modeling Approach to Assess Latent Health Effects
- **Employment:** Research assistant, College of Pharmacy and Nutrition, University of Saskatchewan

[5] **Setu Chandra Kar**, M.Sc., Statistics, Math & Stat, Supervised, 2014–2016 (Defended: 2016)

[4] **Shi Qiu**, M.Sc., Statistics, Math & Stat, Co-supervised with Prof. Cindy Feng, 2012–2015 (Defended: March 26, 2015)

- **Thesis:** Cross-validatory Model Comparison and Divergent Regions Detection using iIS and iWAIC for Disease Mapping
- **Employment:** Statistician, Mabwell Therapeutics, Inc., Shanghai, China (First Employment: Data Service Specialist at IRD Inc.)

[3] **Masud Rana**, M.Sc., Statistics, Math & Stats, Co-supervised with Prof. Shahedul Khan, 2010–2012 (Defended: Sept. 2012)

- **Thesis:** Spatial-Longitudinal Bent-Cable Model with an Application to Atmospheric CFC Data
- **Employment:** Biostatistician at Clinical Research Support Unit, College of Medicine, University of Saskatchewan

[2] **Lai Jiang**, M.Sc., Statistics, Math & Stats, Supervised, 2008–2009 (Transferred to Ph.D.; M.Sc. supervision ended)

[1] **Zhengrong Li**, M.Sc., Statistics, Math & Stats, Supervised, 2007–2012 (Defended: May 2012)

- **Thesis:** A Non-MCMC Procedure for Fitting Dirichlet Process Mixture Models
- **Employment:** Data Service Specialist at IRD Inc.

10.3 Graduate Theses Supervised

2025-2026

[17] Dananji Egodage, 2025, Component-wise Z-residuals for Checking Bayesian Hurdle Models, M.Sc., defended on Aug. 30, 2025.

2024-2025

[16] Effie Wuqian Gao, 2024, Z-residuals for Checking Bayesian Hurdle Models, M.Sc., defended on Aug. 30, 2024.

2023-2024

[15] Tingxuan Wu, 2023, Residual Diagnostics and Statistical Inference for Shared Frailty Models, Ph.D., defended on May 24, 2023.

2022-2023

[14] Hao Hu, 2022, Identifying Risk Factors for Cognitive Decline using Statistical Learning Techniques and Functional Data Analysis, M.Sc., defended on Sept. 15, 2022.

2021-2022

[13] Wutao Yin, 2021, Artificial Intelligence Based Methods for Autism Spectrum Disorder Diagnosis from fMRI Data, Ph.D., defended on Dec. 17, 2021.

[12] Man Chen, 2021, Association Between Gut Microbiome and Parkinson's Disease Revealed by Sparse Learning, M.Sc., defended on April 30, 2021.

2019-2020

[11] Dong Mei, 2019, Feature Selection Bias in Assessing the Predictivity of SNPs for Alzheimer's Disease, M.Sc., defended on May 23, 2019.

[10] Xiaoying Wang, 2019, Comparison of Statistical Testing and Predictive Analysis Methods for Feature Selection in Zero-inflated Microbiome Data, M.Sc., defended on March 12, 2019.

2018-2019

[9] Tingxuan Wu, 2018, Randomized Survival Probability Residuals for Assessing Parametric Survival Models, M.Sc., defended on Dec. 4, 2018.

[8] Wei Bai, 2018, Randomized Quantile Residual for Assessing Generalized Linear Mixed Models with Application to Zero-Inflated Microbiome Data, M.Sc., defended on July 12, 2018.

2016-2017

[7] Yunyang Wang, 2016, Comparison of Stochastic Volatility Models Using Integrated Information Criteria, M.Sc., defended on Nov. 18, 2016.

[6] Alireza Sadeghpour, 2016, Empirical Investigation of Randomized Quantile Residuals for Diagnosis of Non-Normal Regression Models, M.Sc., defended on Sept. 19, 2016.

[5] Naorin Islam, 2016, Substance Abuse and Health: A Structural Equation Modeling Approach to Assess Latent Health Effects, M.Sc., defended on Nov. 28, 2016.

2015-2016

[4] Lai Jiang, 2015, Fully Bayesian T-probit Regression with Heavy-tailed Priors for Selection in High-Dimensional Features with Grouping Structure, Ph.D., defended on Sept. 14, 2015.

2014-2015

[3] Shi Qiu, 2015, Cross-validatory Model Comparison and Divergent Regions Detection using iIS and iWAIC for Disease Mapping, M.Sc., defended on March 26, 2015.

2012-2013

[2] Masud Rana, 2012, Spatial-Longitudinal Bent-Cable Model with An Application to Atmospheric CFC Data, M.Sc., defended in Sept. 2012.

[1] Zhengrong Li, 2012, A Non-MCMC Procedure for Fitting Dirichlet Process Mixture Models, M.Sc., defended in May 2012.

10.4 Supervision of Post-Doctoral Fellows and Research Associates

[4] Tingxuan Wu, University of Saskatchewan, Postdoc Research Associate, Jan. 2025 – July 2026.

[3] Tingxuan Wu, University of Saskatchewan, Postdoc Fellow, June 2023 – April 2024.

[2] Ming Ming Zhang, MITACS Postdoc, 2022-2025.

[1] Jinhong Shi, team-supervised for CFREF projects, Sept. 2016 - Aug. 2019.

10.5 Staff Supervision

- Saima Khosa, faculty mentoring, Sept. 2022- Dec. 2022.

10.6 Thesis Committee Memberships

#	NAME	DEG	PROGRAM	TIME FRAME	ROLE
31	Tiansui Wu	M.Sc.	Biostatistics	2025–Present	Chair
30	Prabhawi Kahatapitiye	Ph.D.	Statistics	2025–Present	Member
29	Rasel Kabir	Ph.D.	Biostatistics	2025–Present	Chair
28	Mohammad Toranjsimin	M.Sc.	Biostatistics	2023–2025 (Def. 09/2025)	Member
27	Lina Li	Ph.D.	Biostatistics	2023–Present	Member
26	Kyle Gardiner	M.Sc.	Statistics	2023–2024 (Def. 09/2024)	Member
25	Mangladeep Bhullar	Ph.D.	Physics	2023–2025 (Def. 11/2025)	Cognate
24	Hammed Jimoh	M.Sc.	Statistics	2022–2023	Member
23	Qi Zhang	M.Sc.	Sociology	2021–2022	External
22	Han Wang	Ph.D.	Sociology	2020–Present	Cognate
21	Yanzhao Cheng	Ph.D.	Biostatistics	2020–2022 (Def. 10/2021)	Chair
20	Naeima Ashleik	Ph.D.	Statistics	2017–2018 (Def. 03/2018)	Member
19	Mehdi Rostami	M.Sc.	Biostatistics	2014–2016 (Def. 06/2016)	Member
18	Sanjeev Rijal	M.Sc.	Statistics	2014–2018 (Def. 07/2017)	Member
17	Farhad Maleki	Ph.D.	Bioinformatics	2014–2019	Cognate
16	Saima Khan Khosa	Ph.D.	Statistics	2014–2017	Member
15	Yue Dong	M.Sc.	Statistics	2014–2016 (Def. 06/2016)	Member
14	Temitope Adesina	M.Sc.	Biostatistics	2014–2015	Chair
13	Sudhakar Achath	M.Sc.	Statistics	2014–2017 (Def. 05/2017)	Member
12	Xiaolei Yu	Ph.D.	Geography	2012–2022 (Def. 06/2022)	Cognate
11	Matthew Schmirler	Ph.D.	Statistics	2012–2022 (Def. 07/2022)	Member
10	Masha Naseri	Ph.D.	Computer Sci.	2012–2014 (Def. 02/2014)	Cognate
9	Chel Hee Lee	Ph.D.	Statistics	2012–2013	Member
8	Weiwei Fan	M.Sc.	Bioinformatics	2012–2014 (Def. 01/2014)	Member
7	Zhaoqin Li	Ph.D.	Geography	2011–2017 (Def. 04/2017)	Cognate

#	NAME	DEG	PROGRAM	TIME FRAME	ROLE
6	Courtney Kendall	M.Sc.	Statistics	2011–2014 (Def. 08/2014)	Member
5	Michael Janzen	Ph.D.	Computer Sci.	2011–2012 (Def. 03/2012)	Cognate
4	Matthew Schmirler	M.Sc.	Statistics	2010–2013 (Def. 09/2012)	Member
3	Mohammed Obeidat	Ph.D.	Statistics	2010–2014 (Def. 07/2014)	Member
2	Lingling Jin	Ph.D.	Bioinformatics	2010–2018 (Def. 08/2017)	Cognate
1	Tolulope Sajobi	Ph.D.	Biostatistics	2008–2012 (Def. 03/2012)	Member

11. BOOKS AND CHAPTERS IN BOOKS

11.1 Authored Books

[2] Soltanifar, M., Li, L., and Rosenthal, J., 2010. A Collection of Exercises in Advanced Probability Theory - the solutions manual of all even-numbered exercises from "A First Look at Rigorous Probability Theory", World Scientific Publishing, (Second Edition, 2006), Singapore.

[1] Li, L., 2007. Bayesian Classification and Regression with High Dimensional Features. (Ph.D. Thesis), Toronto: University of Toronto.

11.3 Chapters in Books

- Feng, C. X. and Li, L ., 2016. Modeling Zero Inflation and Overdispersion in the Length of Hospital Stay for Patients with Ischaemic Heart Disease, in the book Advanced Statistical Methods in Big-Data Sciences, edited by D. Chen, J. Chen, X. Lu, G. Yi and H. Yu, Springer, Chapter 3, pp. 35-53.

12. PAPERS IN REFERRED JOURNALS

2026-2027

[34] Wu, T., Gao, WE, Feng, C., and Li, L., Z-residuals for Diagnosing Bayesian Models, *Journal of American Statistical Association*, under revision.

2025-2026

[33] Wu, T., Li, L., and Feng, C., Z-residuals Diagnostics for Cox Proportional Hazards Models: Distinguishing Functional Form Misspecification from Nonproportional Hazards, with an Application to Biliary Cirrhosis Survival Times, *Canadian Journal of Statistics*, Accepted on June 5, 2026.

[32] Nolan, J., Su, C., Li, L., 2025. Evaluating Railroad Duopoly Behavior: A Market Level Analysis. *Review of Network Economics* 24, 87–111. https://doi.org/10.1515/rne-2025-0034

2024-2025

[31] Wu, T., Feng, C., Li, L., 2025. Cross-validatory Z-Residual for Diagnosing Shared Frailty Models. *The American Statistician*, 79(2), 198–211. https://doi.org/10.1080/00031305.2024.2421370 [**PDF**]; [**Free Reprints**]; [**slides**]; [**Z-residual on Github**]; [**Demo**]

[30] Wu, T., Li, L., Feng, C., 2025. Z-residual diagnostic tool for assessing covariate functional form in shared frailty models. *Journal of Applied Statistics*, 52(1), 28–58. https://doi.org/10.1080/02664763.2024.2355551 [**PDF**]; [**Z-residual on Github**]; [**Demo**] [**slides**]

[29] Wu, T., Feng, C., Li, L., 2025. A Comparison of Estimation Methods for Shared Gamma Frailty Models. *Statistics in Biosciences*, Volume 17, pages 791–812. https://doi.org/10.1007/s12561-024-09444-7 [**PDF**]

2023-2024

[28] Feng, C., Li, L., Xu, C., 2023. Advancements in predicting and modeling rare event outcomes for enhanced decision-making. *BMC Medical Research Methodology* 23, Article 243 (pp. 1-3). https://doi.org/10.1186/s12874-023-02060-x

2021-2022

[27] Yin, W., Li, L., Wu, F.-X., 2022. A semi-supervised autoencoder for autism disease diagnosis. *Neurocomputing*, 483, 140–147. https://doi.org/10.1016/j.neucom.2022.02.017

[26] Cheng, H., Wang, W., Li, L., 2022. Determinants of Citizen Acceptance of White-Collar Crime in China. *Journal of Asian and African Studies*, 59(3), 826-843. https://doi.org/10.1177/00219096221123742 (Published OnlineFirst; final pagination may vary.)

[25] Yin, W., Li, L., Wu, F.-X., 2022. Corrigendum to “Deep learning for brain disorder diagnosis based on fMRI images, Neurocomputing 469 (2022) 332–345”. *Neurocomputing* 509, 271. https://doi.org/10.1016/j.neucom.2022.08.074

[24] Yin, W., Li, L., Wu, F.-X., 2022. Deep learning for brain disorder diagnosis based on fMRI images. *Neurocomputing* 469, 332–345. https://doi.org/10.1016/j.neucom.2020.05.113

2020-2021

[23] Bai, W., Dong, M., Li, L., Feng, C., Xu, W., 2021. Randomized quantile residuals for diagnosing zero-inflated generalized linear mixed models with applications to microbiome count data. *BMC Bioinformatics* 22, Article 564 (pp. 1-17). https://doi.org/10.1186/s12859-021-04371-6 [**slides**].

[22] Li, L., Wu, T., Feng, C., 2021. Model Diagnostics for Censored Regression via Randomized Survival Probabilities. *Statistics in Medicine* 40(6), 1482–1497. https://doi.org/10.1002/sim.8852 [**PDF**]; [**R Functions and Demonstration**]; [**slides**];

[21] Dagasso, G., Yan, Y., Wang, L., Li, L., Kutcher, R., Zhang, W., Jin, L., 2021. Leveraging Machine Learning to Advance Genome-Wide Association Studies. *International Journal of Data Mining and Bioinformatics*, 25(1/2), 17–36. https://doi.org/10.1504/ijdmb.2021.116881

2019-2020

[20] Dong, M., Li, L., Chen, M., Kusalik, A., Xu, W., 2020. Predictive analysis methods for human microbiome data with application to Parkinson’s disease. *PLOS ONE* 15(8), e0237779 (pp. 1-20). https://doi.org/10.1371/journal.pone.0237779

[19] Feng, C., Li, L., Sadeghpour, A., 2020. A comparison of residual diagnosis tools for diagnosing regression models for count data. *BMC Medical Research Methodology* 20, Article 175 (pp. 1-11). https://doi.org/10.1186/s12874-020-01055-2 [**R code used in this paper**]

[18] Jiang, L., Greenwood, C.M.T., Yao, W., Li, L., 2020. Bayesian Hyper-LASSO Classification for Feature Selection with Application to Endometrial Cancer RNA-seq Data. *Scientific Reports* 10, Article 9747 (pp. 1-12). https://doi.org/10.1038/s41598-020-66466-z

[17] Soltanifar, M., Li, L., and Rosenthal, J. S., 2010. A Collection of Exercises in Advanced Probability Theory, World Scientific Publishing, Singapore. [**PDF**].

2018-2019

[16] Shi, J., Yan, Y., Links, M.G., Li, L., Dillon, J.-A.R., Horsch, M., Kusalik, A., 2019. Antimicrobial resistance genetic factor identification from whole-genome sequence data using deep feature selection. *BMC Bioinformatics* 20, Article 535 (pp. 1-14). https://doi.org/10.1186/s12859-019-3054-4

2017-2018

[15] Essien, S. K., Feng, C., Sun, W., Farag, M., Li, L., Gao, Y., 2018. Sleep duration and sleep disturbances in association with falls among the middle-aged and older adults in China: a population-based nationwide study. *BMC Geriatrics* 18, Article 196 (pp. 1-9). https://doi.org/10.1186/s12877-018-0889-x

[14] Li, L., Yao, W., 2018. Fully Bayesian Logistic Regression with Hyper-Lasso Priors for High-dimensional Feature Selection. *Journal of Statistical Computation and Simulation*, 88(14), 2827–2851. https://doi.org/10.1080/00949655.2018.1490418 [**PDF**]; [**software**]; [**slides**].

2016-2017

[13] Jin, L., McQuillan, I., Li, L., 2017. Computational Identification of Harmful Mutation Regions to the Activity of Transposable Elements. *BMC Genomics* 18, Article 862 (pp. 1-10). https://doi.org/10.1186/s12864-017-4227-z

[12] Li, L., Feng, C.X., Qiu, S., 2017. Estimating Cross-validatory Predictive P-values with Integrated Importance Sampling for Disease Mapping Models. *Statistics in Medicine*, 36(14), 2220–2236. https://doi.org/10.1002/sim.7278 [**PDF**]; [**slides**]; [**R Functions**].

[11] Feng, C. X., Rostami, M., Li, L., 2017. Impact of Misspecified Residual Correlation Structure on the Parameter Estimates in a Shared Spatial Frailty Model. *Journal of Statistical Computation and Simulation*, 87(12), 2384–2410. https://doi.org/10.1080/00949655.2017.1332196

2015-2016

[10] Li, L., Qiu, S., Zhang, B., Feng, C.X., 2016. Approximating Cross-validatory Predictive Evaluation in Bayesian Latent Variables Models with Integrated IS and WAIC. *Statistics and Computing*, 26(4), 881–897. https://doi.org/10.1007/s11222-015-9577-2 [**PDF**]; [**slides**].

2013-2014

[9] Yao, W., Li, L., 2014. A New Regression Model: Modal Linear Regression. *Scandinavian Journal of Statistics*, 41(3), 656–671. https://doi.org/10.1111/sjos.12054 [**PDF**].

[8] Yao, W., Li, L., 2014. Bayesian Mixture Labeling by Minimizing Deviance of Classification Probabilities to Reference Labels. *Journal of Statistical Computation and Simulation*, 84(2), 310–323.

2011-2012

[7] Khan, S. A., Rana, M., Li, L., Dubin, J. A., 2012. A Comparative Case Study to Monitor and Understand Atmospheric CFC Decline with the Spatial-Longitudinal Bent-Cable Model. *International Journal of Statistics and Probability*, 1(2), 56–68.

[6] Li, L., 2012. Bias-corrected Hierarchical Bayesian Classification with a Selected Subset of High-dimensional Features. *Journal of American Statistical Association*, 107(497), 120–134. https://doi.org/10.1198/JASA.2011.AP10446 [**PDF**]; [**software**]; [**slides**].

[5] Sajobi, T.T., Lix, L. M., Dansu, B. M., Laverty, W., Li, L., 2012. Robust Descriptive Discriminant Analysis for Repeated Measures Data. *Computational Statistics & Data Analysis*, 56(9), 2782–2794. https://doi.org/10.1016/j.csda.2012.02.029

2010-2011

[4] Sajobi, T. T., Lix, L., Li, L., Laverty, W., 2011. Discriminant Analysis for Repeated Measures Data: Effects of Mean and Covariance Misspecification on Bias and Error in Discriminant Function Coefficients. *Journal of Modern Applied Statistical Methods*, 10(2), 571–582. https://doi.org/10.22237/jmasm/1320120840

2009-2010

[3] Li, L., 2010. Are Bayesian Inferences Weak for Wasserman's Example? *Communications in Statistics – Simulation and Computation*, 39(4), 655–667. https://doi.org/10.1080/03610910903576540 [**PDF**]; [**slides**].

2007-2008

[2] Li, L., Zhang, J., Neal, R.M., 2008. A method for avoiding bias from features selection with application to naive Bayes classification models. *Bayesian Analysis*, 3(1), 171–196. https://doi.org/10.1214/08-BA307 [**PDF**]; [**slides**]; [**software**].

[1] Li, L., Neal, R.M., 2008. Compressing Parameters in Bayesian High-order Models with Application to Logistic Sequence Models. *Bayesian Analysis*, 3(4), 793–822. https://doi.org/10.1214/08-BA330 [**PDF**]; [**slides**]; [**software**].

13. REFERRED CONFERENCE PUBLICATIONS

[3] Yin, W., Li, L., Wu, F.-X., 2021. A Graph Attention Neural Network for Diagnosing ASD with fMRI Data, in: 2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). pp. 1131–1136. https://doi.org/10.1109/BIBM52615.2021.9669849

[2] Dagasso, G., Yan, Y., Wang, L., Li, L., Kutcher, R., Zhang, W., Jin, L., 2020. Comprehensive-GWAS: a pipeline for genome-wide association studies utilizing cross-validation to assess the predictivity of genetic variations, in: 2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM). Presented at the 2020 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp. 1361–1367. https://doi.org/10.1109/BIBM49941.2020.9313355

[1] Jin, L., McQuillan, I., and Li, L., 2016, Computational Identification of Regions that Influence Activity of Transposable Elements in the Human Genome. Proceeding of 2016 IEEE International Conference on Bioinformatics and Biomedicine, pp. 592-599.

14. PRESENTATIONS

14.1 Invited Presentations

2025-2026

[33] Z-residuals for Checking Bayesian Models. Presented at: University of Calgary, Calgary, AB, Canada; July 28, 2025

[32] Sparse Learning for Assessing the Association Between Gut Microbiome and Parkinson's Disease. Presented at: The 3rd JCSDS, Hangzhou, China, July 13, 2025.

2024-2025

[31] Z-residuals for Checking Bayesian Models. Presented at: International Conference on Statistics and Data Science, Vancouver, BC, Canada; June 24, 2025

[30] Z-residuals for Checking Bayesian Hurdle Models. Presented at: EcoStat 2024; July 17, 2024; Beijing, China.

2023-2024

[29] Z-residual Diagnostic Tool for Assessing Covariate Functional Form in Proportional Hazards Models with Shared Frailty, ICSA Canada Chapter Symp., June 9, 2024, Niagara Falls, Canada

[28] Z-residual Diagnostic Tool for Assessing Covariate Functional Form in Proportional Hazards Models with Shared Frailty, Annual Meeting of SSC, St John's, Canada, June 2, 2024

[27] Z-residual Diagnostic Tool for Assessing Covariate Functional Form in Proportional Hazards Models with Shared Frailty, Dept. Seminar, Texas State University, USA, March 8, 2024

[26] Z-residual Diagnostic Tool for Assessing Covariate Functional Form in Proportional Hazards Models with Shared Frailty, Dept. Seminar, Sun Yat-sen University, China, Jan. 4, 2024

2022-2023

[25] Cross-validatory Residual Diagnostics for Bayesian Spatial Models, Annual Meeting of SSC, Ottawa, May 29, 2023

[24] Model Diagnostics for Censored Regression via Randomized Survival Probabilities, the 5th ICSA Canada Symposium, 9 July 2022, Banff, AB, Canada

[23] Model Diagnostics for Censored Regression via Randomized Survival Probabilities, 17 Aug. 2022, Statistics Conference in Genomics, Pharmaceutical Science, and Health Data Science, University of Victoria, Victoria, BC, Canada

2021-2022

[22] Randomized quantile residuals for diagnosing zero-inflated generalized linear mixed models with applications to microbiome count data, SSC Annual Meeting (virtual), May 2022.

[21] Model Diagnostics for Censored Regression via Randomized Survival Probabilities, The 6th Canadian Conference in Applied Statistics, Hosted by Concordia University (virtual), 16 July 2021.

2019-2020

[20] Estimating Cross-validatory Predictive P-values with Integrated Importance Sampling for Disease Mapping Models, Aug. 2019, the 4th ICSA-Canada Symposium held at Queen's University.

2018-2019

[19] Feature Selection Bias in Assessing the Predictivity of SNPs for Alzheimer's Disease, June 2019, Seminar talk, University of Manitoba, Canada

2017-2018

[18] Randomized Quantile Residuals for Checking GLMM with Application to Zero-inflated Microbiome Data, June 2018, Annual Meeting of Statistical Society of Canada, McGill University, Canada.

[17] Fully Bayesian Classification with Heavy-tailed Priors for Selection in High-Dimensional Features with Grouping Structure, Aug. 2017, the 3rd ICSA-Canada Symposium held at Vancouver.

2016-2017

[16] Randomized Quantile Residuals: an Omnibus Model Diagnostic Tool with Unified Reference Distribution, June 2017, Seminar talk, School of Mathematical Sciences, Xiamen University, China.

[15] Fully Bayesian Classification with Heavy-tailed Priors for Selection in High-Dimensional Features with Grouping Structure, June 2017, Seminar talk, School of Mathematical Sciences, Xiamen University, China.

[14] Randomized Quantile Residuals: an Omnibus Model Diagnostic Tool with Unified Reference Distribution, June 2017, Seminar talk, Department of Biostatistics, Southern Medical University, Guangzhou, China.

[13] Estimating Cross-validatory Predictive P-values with Integrated Importance Sampling for Disease Mapping Models, June 2017, Annual Meeting of Statistical Society of Canada, University of Manitoba, Canada.

[12] Fully Bayesian Classification with Heavy-tailed Priors for Selection in High-Dimensional Features with Grouping Structure, Dec., 2016, Wuhan University, China.

2015-2016

[11] Cross-validatory Model Comparison and Divergent Regions Detection using iIS for Disease Mapping, Seminar of Dept of Math & Stat, University of Calgary, April 2016, Calgary, AB.

[10] Cross-validatory Model Comparison and Divergent Regions Detection using iIS for Disease Mapping, Seminar of Dept of Math & Stat, University of Alberta, Edmonton, AB.

[9] Cross-validatory Model Comparison and Divergent Regions Detection using iIS for Disease Mapping, Seminar of Department of Statistics, University of Manitoba, Jan. 2016, Winnipeg, MB.

[8] Bias-corrected Hierarchical Bayesian Classification with a Selected Subset of High-dimensional Features, ICSA Canada Chapter Annual Meeting, University of Calgary, Aug. 2015, Calgary, AB.

2014-2015

[7] Approximating Cross-validatory Predictive Evaluation in Bayesian Latent Variables Models with Integrated IS and WAIC, Dec. 2014, Tongji University, Shanghai, China.

[6] An Introduction to Microarray Data. Workshop on "Statistical Issues in Biomarker and Drug Co-development", Nov. 2014, Fields Institute, Toronto, ON, Canada.

2013-2014

[5] Approximating Cross-validatory Predictive Evaluation in Bayesian Latent Variables Models with Integrated IS and WAIC. Statistics Seminar, April, Kansas State University, Manhattan, Kansas, USA.

2011-2012

[4] High-dimensional Feature Selection Using Hierarchical Bayesian Logistic Regression with Heavy-tailed Priors. CRM-ISM-GERAD Colloque de Statistique, April, McGill University, Montreal, Quebec, Canada.

2010-2011

[3] High-dimensional Classification using Hierarchical Bayesian Polychotomous Logistic Regression Models. Colloquia talk, Jan., The University of Western Ontario, London, ON, Canada.

[2] High-dimensional Classification using Hierarchical Bayesian Polychotomous Logistic Regression Models. Colloquia talk, Sept., Penn State University, University Park, PA, USA.

2007-2008

[1] Avoiding Bias from Feature Selection. CRISM 'workshop on Bayesian Analysis of High-dimensional Data, April, University of Warwick, Coventry, UK.

14.2 Contributed Presentations

2013-2014

[10] Approximating Cross-validatory Predictive Evaluation in Bayesian Latent Variables Models with Integrated IS and WAIC. Annual Meeting of Statistical Society of Canada, May 27, 2014, Toronto, ON, Canada.

2010-2011

[9] High-dimensional Classification using Hierarchical Bayesian Polychotomous Logistic Regression Models. The 8th ICSA International Conference, Dec. 20, 2010, Guangzhou, China.

2009-2010

[8] Sajobi, T., Lix, L., Laverty, W., and Li, L., 2010. Discriminant Analysis for Repeated Measures Data: Effects of Covariance Structure on Bias and Error in Discriminant Function Coefficients. Annual Meeting of Statistical Society of Canada, May 24, 2010, Quebec City, QC, Canada.

[7] Are Bayesian Inferences Weak for Wasserman's Example? Annual Meeting of Statistical Society of Canada, May 25, 2010, Quebec City, QC, Canada.

2008-2009

[6] Calibrating Predictions Based on a Selected Subset of Features from Bayesian Gaussian Classification Models. Annual meeting of Statistical Society of Canada, January, Vancouver, BC, Canada.

[5] Calibrating Predictions Based on a Selected Subset of Features from Bayesian Gaussian Classification Models. Bayesian Biostatistics Conference, January, Houston, TX, USA.

2007-2008

[4] Compressing Parameters in Bayesian High-order Models. Annual Meeting of Statistical Society of Canada, May, Ottawa, ON, Canada.

2006-2007

[3] Compressing Parameters in Bayesian Models with High-order Interactions. The 3rd Monte Carlo Workshop, Harvard University, May, Cambridge, MA, USA.

2005-2006

[2] Avoiding Bias from Feature Selection in Regression and Classification Models. Joint Statistical Meeting, August, Seattle, WA, USA.

[1] Analysis of Obstructive Sleep Apnea Data with Bayesian Neural Network. Annual Meeting of Statistical Society of Canada, June, London, ON, Canada.

15. REPORTS AND OTHER OUTPUTS

15.1 Software Released Publicly

[11] Wu, T. and Li, L., 2026. `Zresidual`: Computing and Diagnosing Gaussian-like Residuals. [Github] [Demo]. Version 0.1-0 on Github (2026).

[10] Li, L., 2026. R Functions for Computing Z-residuals for `survreg` and `coxph` Objects. [URL].

[9] Li, L., et al., 2021. Real-time estimates of R_t for Covid-19 in Canada. [URL].

[8] Li, L. and Liu, S., 2019–2026. `HTLR`: Bayesian Logistic Regression with Hyper-LASSO priors. DOI: 10.32614/CRAN.package.HTLR. [CRAN] [Github] [URL]. Version 0.4 (2019), version 0.4-1 (2019), version 0.4-2 (2020), version 0.4-3 (2020), version 0.4-4 (2022), version 1.0 (2026).

[7] Li, L., 2011–2026. `BCBCSF`: Bias-corrected Bayesian Classification with Selected Features. DOI: 10.32614/CRAN.package.BCBCSF. [CRAN] [URL]. Version 0.0-0 (2011), version 0.0-1 (2011), version 0.0-2 (2012), version 1.0-0 (2013), version 1.0-1 (2015), updated to version 1.0-2 (2026).

[6] Li, L., 2018. `HTLR`: Bayesian Logistic Regression with Hyper-LASSO priors. [URL]. Pre-CRAN version (2018).

[5] Li, L., 2016. `iIS`: R code for computing predictive p-values in disease mapping models. [URL].

[4] Li, L., 2008. `gibbs.met`: Naive Gibbs Sampling with Metropolis Steps. [CRAN] [URL].

[3] Li, L., 2008. `BPHO`: Bayesian Prediction with High-order Interactions. [CRAN] [URL].

[2] Li, L., 2007. `predmixcor`: Classification rule based on Bayesian mixture models with feature selection bias corrected. [CRAN] [URL].

[1] Li, L., 2007. `predbayescor`: Classification Rule Based on Bayesian Naive Bayes Models with Features Selection Bias Corrected. [CRAN] [URL].

15.2 Technical Reports

[12] Wu, T., Feng, C. and Li, L., 2023. Cross-validatory Z-Residual for Diagnosing Shared Frailty Models. https://doi.org/10.48550/arXiv.2303.09616. 32 pages, 14 figures.

[11] Wu, T., Li, L. and Feng, C., 2023. Z-residual diagnostics for detecting misspecification of the functional form of covariates for shared frailty models. https://doi.org/10.48550/arXiv.2302.09106. 21 pages, 7 figures.

[10] Li, L., Wu, T. and Feng, C., 2019. Model diagnostics for censored regression via randomized survival probabilities. https://doi.org/10.48550/arXiv.1911.00198. 12 pages. (Journal-ref: Statistics in Medicine, 2021, 40(6), 1482-1497).

[9] Feng, C., Sadeghpour, A. and Li, L., 2017. Randomized Predictive P-values: A Versatile Model Diagnostic Tool with Unified Reference Distribution. https://doi.org/10.48550/arXiv.1708.08527. 26 pages. (Journal-ref: BMC Medical Research Methodology, 2020, 20(175)).

[8] Jiang, L., Li, L. and Yao, W., 2016. Fully Bayesian Classification with Heavy-tailed Priors for Selection in High-dimensional Features with Grouping Structure. https://doi.org/10.48550/arXiv.1607.00098. 31 pages. (Journal-ref: Sci Rep, 2020, 10(9747)).

[7] Li, L., Feng, C.X. and Qiu, S., 2016. Estimating Cross-validatory Predictive P-values with Integrated Importance Sampling for Disease Mapping Models. https://doi.org/10.48550/arXiv.1603.07668. 18 pages. (Journal-ref: Statistics in Medicine, 2017, 36(14), 2220-2236).

[6] Li, L. and Yao, W., 2014. Fully Bayesian Logistic Regression with Hyper-Lasso Priors for High-dimensional Feature Selection. https://doi.org/10.48550/arXiv.1405.3319. 33 pages. (Journal-ref: Journal of Statistical Computation and Simulation, 2018, 88(14), 2827-2851).

[5] Li, L., Qiu, S., Zhang, B. and Feng, C.X., 2014. Approximating Cross-validatory Predictive Evaluation in Bayesian Latent Variables Models with Integrated IS and WAIC. https://doi.org/10.48550/arXiv.1404.2918. 38 pages. (Journal-ref: Statistics and Computing, 2016, 26(4), 881-897).

[4] Li, L. and Yao, W., 2013. High-dimensional Feature Selection Using Hierarchical Bayesian Logistic Regression with Heavy-tailed Priors. https://doi.org/10.48550/arXiv.1308.4690. (Earlier version of arXiv:1405.3319).

[3] Li, L. and Neal, R.M., 2007. A Method for Compressing Parameters in Bayesian Models with Application to Logistic Sequence Prediction Models. https://doi.org/10.48550/arXiv.0711.4983. 29 pages. (Journal-ref: Bayesian Analysis, 2008, 3(4), 793-822).

[2] Li, L., 2007. Bayesian Classification and Regression with High Dimensional Features. https://doi.org/10.48550/arXiv.0709.2936. PhD Thesis Submitted to University of Toronto, 129 pages.

[1] Li, L., Zhang, J. and Neal, R.M., 2007. A method for avoiding bias from features selection with application to naive Bayes classification models. Technical Report No 0705, Department of Statistics, University of Toronto.

15.3 Online Apps

[5] **Students' Grade Calculator:** A Shinylive app that can calculate students' grades with fine-grained controls and output.

[4] **Animation of Finding $\sqrt{S}$:** A Shinylive App for Finding Square Root using Newton Method

[3] **Abbreviation Extractor for Documents with Latex Equations:** A shinylive app that can extract abbreviations in the source text with latex equations.

[2] Real-time estimates of the reproduction rate (R_t) of Canada and its provinces, a website maintained until Feb 2022.

[1] Projections of Canada's COVID-19 Cases (up to 2020-06-30)

17. RESEARCH FUNDING HISTORY

2025-2026

[17] **NSERC Individual Discovery Grant (No. 2026-07053)** – *Prediction-based Methods for Statistical Learning and Inference in Biosciences and Epidemiology*, $185,000 (37K per year), 2026-2031, PI.

[16] **CANSSI** – *Statistical Methodologies and Computational Tools to Identify Microbial Correlates of Canadian Bee Gut Health*, Collaborative Research Team Projects – Project 29, 2025-2028, Co-PI.

2021-2022

[15] **MITACS Accelerate Grant** – *Geospatial Artificial Intelligence Algorithms for Automating Manual Observation Associated with Wheat Production*, $280,000, 2021-2025, PI.

2020-2021

[14] **MITACS Accelerate Grant** – *Develop a web-based geospatial artificial intelligence framework to track, visualize, analyze, model, and predict infectious disease spread in real-time*, $105,000, 2020-2021, PI.

2019-2020

[13] **NSERC Individual Discovery Grant** – *Predictive Methods for Analyzing High-throughput and Spatial-temporal Data*, $140,000 (20K per year), 2019-2026, PI.

2017-2018

[12] **The Western Canadian Universities Collaborative Project Seed Funding** – *Genome-wide diet-gene interaction analysis for risk of psychiatric comorbidity in inflammatory bowel disease*, $20,000, 2017-2019, Co-PI.

2016-2017

[11] **Canada First Research Excellence Fund (CFREF)** - *Designing Crops for Global Food Security, Genotype & Environment to Phenotype*, $756,918, 2016-2019, Co-Investigator (PI: Prof. Kusalik).

[10] **MITACS Accelerate Internship** – *Applications of Neural Network Curve Fitting Methods for Least-squares Monte Carlo Simulations in Financial Risk Management*, $15,000, 2016, PI.

2014-2015

[9] **NSERC Individual Discovery Grant** – *Bayesian Methods for High-dimensional and Correlated Data*, $70,000, 2014-2019, PI.

2011-2012

[8] **NSERC Individual Discovery Grant ECR Supplement** – *Efficient Bayesian Analysis for Complex Models*, $5,000/year, 2011-2014, PI.

2009-2010

[7] **NSERC Individual Discovery Grant** – *Efficient Bayesian Analysis for Complex Models*, $80,000, 2009-2014, PI.

[6] **CFI Leaders Opportunity Funds** – *A Computer Cluster for Research on Efficient Bayesian Statistical Methods*, $160,000, 2009, PI.

2008-2009

[5] **MITACS Accelerate Internship** – *Clustering Analysis for Detecting the Types of Vehicles*, $15,000, 2008, Co-PI with Prof. Laverty.

[4] **University of Saskatchewan President's Award**, $5,000, 2008, PI.

[3] **College of Graduate Studies and Research at the University of Saskatchewan Award**, $15,000, 2008, PI.

[2] **College of Arts and Science at the University of Saskatchewan – Supplemental start-up operating grant**, $15,000, 2008, PI.

2007-2008

[1] **University of Saskatchewan – Start-up operating grant**, $5,000, 2007, PI.

18. PRACTICE OF PROFESSIONAL SKILLS

2026-2027

[70] Refereeing for *Journal of Computational and Graphical Statistics*, July 2026

[69] Refereeing for *Bioinformatics*, July, 2026

2025-2026

[68] Refereeing for *Journal of Statistical Computation and Simulation*, June, 2026

[67] Refereeing for *Journal of Statistical Computation and Simulation*, April, 2026

[66] Refereeing for *Journal of the Royal Statistical Society: Series C*, April 2026

[65] Refereeing for *Bioinformatics*, March 2026

[64] Refereeing for *Journal of Computational and Graphical Statistics*, March 2026

[63] External Referee for a tenure and promotion of SFU, Dec. 2025

[62] External Examiner for the doctoral thesis by Xiaoqing Zhang at University of Regina, Dec. 8, 2025

[61] Refereeing for *Journal of Statistical Computation and Simulation*, Dec. 2025

[60] Refereeing for *Journal of Computational and Graphical Statistics*, Sept. 2025

[59] External Examiner for the doctoral thesis by Na Zhang at University of Alberta, August 28, 2025

[58] Refereeing for *Journal of the Royal Statistical Society: Series C*, August 2025

[57] Refereeing for *Journal of Applied Statistics*, August 2025

2024-2025

[56] Organizing an invited Session for the 7th Symposium of ICSA Canada Chapter, McGill University, August 2026

[55] Organizing an invited Session for 2025 SSC Annual Meeting, Saskatoon, SK, Canada, June 2025

2023-2024

[54] External Examiner for the doctoral thesis by Yuping Yang at SFU, June 25, 2024

[53] Refereeing for *Journal of Computational and Graphical Statistics*, Sept. 2024

[52] Refereeing for *Journal of Applied Statistics*, Jan. 2024

[51] Review an MITACS Accelerate Grant, Dec. 2023

2022-2023

[50] Refereeing for *Statistical Methods in Medical Research*, April 2023

[49] Refereeing for *Statistical Methods in Medical Research*, Jan. 2023

[48] Refereeing for *Journal of Computational and Graphical Statistics*, Jan. 2023

[47] Refereeing for *Statistical Methods in Medical Research*, Aug. 2022

[46] External Examiner for the M.Sc. thesis by Xiangling Ji, University of Victoria, 27 July 2022

[45] Refereeing for *Canadian Journal of Statistics*, July 2022

2021-2022

[44] Organizer of an invited session for ICSA Canada Symposium 2022, Banff, AB, Canada, July 2022

[43] External Examiner for one NSERC IDG application

[42] External Examiner for another NSERC IDG application

[41] External Examiner for one MITACS Accelerate grant application

[40] Reviewer for a Canada Research Chair Position

[39] External M.Sc. Thesis Examiner for Zhongyuan Zhang, University of Toronto

[38] Refereeing for *Statistical Methods in Medical Research*

[37] Refereeing for *Journal of Statistical Computation and Simulation*

[36] Refereeing for *BMC Cancer*

[35] Refereeing for *Journal of Computational and Graphical Statistics*

[34] Refereeing for *Canadian Journal of Statistics*

[33] Refereeing for *IEEE Transactions on Neural Networks and Learning Systems*

2020-2021

[32] Grant refereeing for an application to MITACS Accelerate, May 2021

[31] External Examination for a PhD thesis of University of Montreal, May 2021

[30] Grant refereeing for an application to NSERC IDG, Jan. 2021

[29] Refereeing for *Statistics in Medicine*

[28] Refereeing for *Computational Statistics and Data Analysis*

[27] Refereeing for *Frontiers in Genetics*

[26] Refereeing for *Statistical Methods for Medical Research*

[25] Refereeing for *Journal of Statistical Computation and Simulation*

[24] Refereeing for *BMC Cancer*

2019-2020

[23] External doctoral thesis examiner for Shijia Wang, Simon Fraser University

[22] External doctoral thesis examiner for Kexin Luo, Western University

[21] Grant Refereeing for an application to MITACS

[20] Grant Refereeing for an application to NSERC IDG

[19] Refereeing for *Computational Statistics and Data Analysis*

[18] Refereeing for *Frontiers in Genetics*

[17] Refereeing for *Communications in Statistics - Simulation and Computation*

2017-2018

[16] Referee a NSERC discovery grant application

[15] Refereeing for *Canadian Journal of Statistics*

[14] Refereeing for *Journal of Royal Statistical Society (C)*

2016-2017

[13] Referee two applications for MITACS Accelerate Grant

[12] Refereeing for *Statistics in Medicine*

[11] Refereeing for *Statistics and Computing*

[10] Refereeing for *PLOS ONE*

2015-2016

[9] Organizer, Invited session "Recent Advances in Statistical Inference Methods in Regression Models for Complex and Big Data", China Statistics Conference, June 2016, Qingdao, China

[8] Refereeing application for NSERC individual discovery grant (2015)

2013-2014

[7] Reviewing and revision services for an SHRF Establishment Grant application (Prof. Kelly Penz), funded 2013

[6] Refereeing for *Biometrika*

[5] Refereeing for *Statistics In Medicine*

[4] Refereeing for *Statistical Papers*

[3] Refereeing for *Computational Statistics*

[2] Refereeing for *Statistica Sinica*

2011-2012

[1] Refereeing application for NSERC individual discovery grant (2011)

19. ADMINISTRATIVE SERVICE

19.1 University Committees

2026-2027

[11] Chair, Collaborative Biostatistics Program, University of Saskatchewan.

2025-2026

[10] Chair, Collaborative Biostatistics Program, University of Saskatchewan.

[9] USASK NSERC Discovery Grant (DG), Internal Reviewer.

2020-2021

[8] USASK NSERC Discovery Grant (DG), Internal Reviewer.

2019-2020

[7] Chair, Collaborative Biostatistics Program, University of Saskatchewan.

2018-2019

[6] Member, Academic Programming Committee, University of Saskatchewan.

2017-2018

[5] Member, Academic Programming Committee, University of Saskatchewan.

2016-2017

[4] Member, Academic Programming Committee, University of Saskatchewan.

[3] Member, University of Saskatchewan Bioinformatics Program Committee.

2015-2016

[2] Member, University of Saskatchewan Bioinformatics Program Committee.

2013-2014

[1] Dean's Designate for the Ph.D. Defense of Rui Zhang, Department of Veterinary Microbiology, June 12, 2014.

19.2 College and Departmental Committees

2025-2026

[45] Member, Budgeting and Planning Committee, Dept of Math & Stat

[44] Co-Chair, Undergraduate Committee (Statistics), Dept of Math & Stat

[43] Member, Graduate Program Committee in Statistics, Dept of Math & Stat

2024-2025

[42] Member, Graduate Program Committee in Statistics, Dept of Math & Stat

[41] Search subcommittee for a faculty position in statistics

2023-2024

[40] Member, Graduate Program Committee in Statistics, Dept of Math & Stat

[39] Member, Sub Search Committee, 4-Year Lecturer Position

2022-2023

[38] Statistics Advisor (credit transferring for the whole university)

[37] Member, Department Promotion (Associate) Committee (1-Case), Dept of Math & Stat

[36] Member, Department Renewals and Tenure Committee (1-Case: Tenure), Dept of Math & Stat

[35] Member, Department Promotion (Full) Committee (1-Case), Dept of Math & Stat

2021-2022

[34] Statistics Advisor (credit transferring for the whole university)

[33] Member, Department Renewals and Tenure Committee (1-Case), Dept of Math & Stat

[32] Member, Department Promotion (Full) Committee (1-Case), Dept of Math & Stat

[31] Search subcommittee for a 4-year term lecturer position (two rounds of searching, Jan 2022–June 2022)

2019-2020

[30] Member, Tenure Committee (1 case), Dept of Math & Stat

[29] Member, Promotion Committee (1 case), Dept of Math & Stat

[28] Member, Renewal of Probation Committee (1 case), Dept of Math & Stat

[27] Member, Graduate Committee, Dept of Math & Stat

[26] Organizer, Team discussion towards renovating undergraduate statistics program, Dept of Math & Stat

2018-2019

[25] Committee Member, Data Science Boot Camp, University of Saskatchewan (June 10–21, 2019)

[24] Member, Curriculum Renewal Committee, Dept of Math & Stat, Term 1

[23] Member, Undergraduate Committee, Dept of Math & Stat, Term 1

2017-2018

[22] Member, Salary Review Committee, Dept of Math & Stat, University of Saskatchewan

[21] Member, Search Committee, Dept of Math & Stat, University of Saskatchewan

2016-2017

[20] Member, Sub Search Committee for a joint position in ``data science/big data'', College of Arts and Science, University of Saskatchewan.

[19] Member, Graduate Committee, Dept of Math & Stat

[18] Member, Sub Search Committee, APA position, Dept of Math & Stat, University of Saskatchewan

[17] Member, Sub Search Committee, 4 lecturer positions, Dept of Math & Stat, University of Saskatchewan

[16] Organizer, Statistics and Probability Alumni Networking Day (Nov. 2016)

[15] Organizer, Qualifying Exams for Trisha Lawrence (Nov. 2016)

2015-2016

[14] Member, Graduate Committee, Dept of Math & Stat

[13] Organizer, Student Seminar Day, Dept of Math & Stat (May 2016)

[12] Team leader, submission of U of S courses for accreditation by the Statistical Society of Canada (May 2016)

[11] Organizer, Qualifying Exams for Trisha Lawrence (May 2016)

2014-2015

[10] Member, Academic Program Committee, College of Arts and Science.

[9] Member, Graduate Committee, Dept of Math & Stat

[8] Organizer, Seminar Series, Dept of Math & Stat

2012-2013

[7] Member, Curriculum Renewal Committee, Dept of Math & Stat

[6] Member, Budget Planning Committee, Dept of Math & Stat

[5] Member, Colloquium Committee, Dept of Math & Stat

2011-2012

[4] Member, Salary Review Committee, Dept of Math & Stat, University of Saskatchewan

[3] Organizer, Seminar Series, Dept of Math & Stat

[2] Member, Colloquium Committee, Dept of Math & Stat

2009-2010

[1] Member, Sub Search Committee, Dept of Math & Stat

20. PROFESSIONAL OR ASSOCIATION OFFICES AND COMMITTEE ACTIVITY

2024-2025

[9] Member of NSERC Discovery Grant EG 1508 Committee

[8] Local Organizing Committee, 2025 Annual Meeting of the Statistical Society of Canada held at the U of S

2023-2024

[7] Member of NSERC Discovery Grant EG 1508 Committee

2022-2023

[6] Member of NSERC Discovery Grant EG 1508 Committee

[5] Co-editor for a special issue “Prediction Methods for Rare Diseases or Outcomes” in the journal *BMC Medical Research Methodology*

2021-2022

[4] Member, CANSSI-SK Health Research Collaborating Center. Participate Substantially in organizing a semester-long seminar series

2019-2020

[3] Program Committee Member for the 4th ICSA-Canada Symposium, Queens University (Aug. 2019)

2017-2018

[2] Co-chair of the scientific program, the 3rd ICSA Canada Chapter Symposium held in Vancouver (Aug. 2017)

2016-2017

[1] Judge for case study competition, Annual Meeting of Statistical Society of Canada (June 2017)